GARAGE MANAGEMENT SYSTEM

SYSTEM DOCUMENTATION

SUBMITTED BY,

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A SYSTEM DOCUMENTATION SUBMITTED IN PARTIAL FULFILMENT FOR THE AWARD OF DIPLOMA IN S0FTWARE ENGINEERING BY ZETECH UNIVERSITY

APRIL 2026

DECLARATION

I, JULIUS NJOROGE**,** declare that this project titled “Garage Management System” is my original work and to the best knowledge it has not been submitted elsewhere for certification.

Student Name: _____ Sign: _____ Date: _____

Supervisor Name: _____ Sign: _____ Date: _____

DEDICATION

I dedicate this project to my family, lecturer, friends, and to everyone who supported me throughout my academic journey.

ABSTRACT

This project presents the development of a Garage Management System designed to improve operations in automotive service environments. The study is structured into three main chapters: Chapter One covers project analysis and planning, including system requirements and objectives; Chapter Two focuses on system design, outlining the architecture and database structure; and Chapter Three describes system implementation and development.

The system aims to enhance efficiency and productivity by automating key processes such as customer management, job tracking, and order processing. It reduces reliance on manual methods, thereby minimizing errors and improving data accuracy and reliability. In addition, the system enhances customer experience through faster service delivery and better record management.

The system was developed using HTML, CSS, JavaScript, PHP, and MySQL to create a dynamic and user-friendly platform. The Waterfall Model was adopted as the development methodology to ensure a structured and systematic approach. Overall, the proposed system provides a more efficient and reliable solution compared to traditional manual garage management practices.

DEFINITION OF KEY TERMS

Admin (Administrator):
A system user with full access rights responsible for managing users, services, pricing, reports, and system configurations.

Billing:
The process of generating invoices and managing customer payments for services rendered.

Customer:
An individual who owns a vehicle and requests repair or maintenance services from the garage.

Database:
An organized collection of data stored electronically and accessed by the system for record keeping and reporting.

Garage Management System (GMS):
A computerized system designed to manage and automate garage operations such as customer records, vehicle services, job tracking, inventory, billing, and reporting.

Inventory Management:
The process of recording, monitoring, and controlling spare parts and materials used in garage operations.

Job Tracking:
A system feature that monitors the progress of vehicle repairs from initiation to completion.

Mechanic:
A garage employee responsible for diagnosing, repairing, and updating the status of assigned vehicle service jobs.

Receptionist:
A staff member responsible for registering customers and vehicles, scheduling appointments, and managing payments.

Service Scheduling:
The process of planning and assigning service jobs to mechanics based on availability and customer appointments.

User Interface (UI):
The visual part of the system through which users interact with the application.

ABBREVIATIONS AND ACRONYMS

GMS - Garage Management System

UI - User Interface

IDE - Integrated Development Environment

HTML - HyperText Markup Language

CSS - Cascading Style Sheets

PHP - Hypertext Preprocessor

SQL - Structured Query Language

DBMS - Database Management System

HDD - Hard Disk Drive

SSD - Solid State Drive

LIST OF FIGURES

Figure 2.2. 1 Main Page 6
Figure 2.2. 2 Admin Dashboard 7
Figure 2.2. 3 Receptionist Dashboard 7
Figure 2.2. 4 Mechanic Dashboard 8
Figure 2.2. 5 Customer Dashboard 8
Figure 2.2. 6 Login/Registration form 9
Figure 2.2. 7 Vehicle Registration Form 9
Figure 2.2. 8 Appointment Registration Form 10
Figure 2.2. 9 Service Form 10
Figure 2.2. 10 Inventory Form 11

Figure 2.3. 1 Login Flowchart 11
Figure 2.3. 2 Use Case Diagram 12
Figure 2.3. 3 Entity Relationship Diagram 13
Figure 2.3. 4 Dfd Level 0 13
Figure 2.3. 5 Dfd Level 1 14
Figure 2.3. 6 Dfd Level 2 14
Figure 2.3. 7 vehicle Registration Dataflow 15
Figure 2.3. 8 Appointment Booking Dataflow 15
Figure 2.3. 9 Inventory Dataflow 16
Figure 2.3. 10 Service Dataflow 16

Figure 3.3. 1 Main Page 17
Figure 3.3. 2 Main Page source code 19
Figure 3.3. 3 Admin Dashboard 20
Figure 3.3. 4 Admin Dashboard Source Code 21
Figure 3.3. 5 Admin Inventory 21
Figure 3.3. 6 Admin Inventory Source Code 23
Figure 3.3. 7 Admin job scheduling 23
Figure 3.3. 8 Admin Job Scheduling Source Code 24
Figure 3.3. 9 Admin users management 25
Figure 3.3. 10 Admin Users Management Source Code 26
Figure 3.3. 11 Admin service management 27
Figure 3.3. 12 Admin Service Management Source Code 28
Figure 3.3. 13 Customer appointment booking 28
Figure 3.3. 14 Customer Appointment Booking Source Code 30
Figure 3.3. 15 Mechanic inventory usage 30
Figure 3.3. 16 Mechanic Inventory Usage Source Code 32
Figure 3.3. 17 Receptionist vehicle Registration 32
Figure 3.3. 18 Receptionist Vehicle Registration Source Codes 33

LIST OF TABLES

Table 1.4 Functional Requirements 3
Table 2.6 Project Schedule Breakdown 5
Table 3.4.1 Issues and Fixes 34

TABLE OF CONTENTS

Contents

DECLARATION i
DEDICATION ii
ABSTRACT iii
DEFINITION OF KEY TERMS iv
ABBREVIATIONS AND ACRONYMS vi
LIST OF FIGURES vii
LIST OF TABLES viii
TABLE OF CONTENTS ix
CHAPTER ONE: PROJECT PLANNING AND ANALYSIS 1
1.1 Statement of Problem 1
1.2 Study Justification 1
1.3 System Objectives 2
1.3.1 General Objective 2
1.3.2 Specific Objectives 2
1.4 Functional Requirements 2
1.5 Breakdown of Tools & Resources to Be Used 3
1.5.1 Laptop 3
1.5.2 Front End 3
1.5.3 IDE (Integrated Development Environment) 4
1.5.4 Back End 4
1.5.5 Internet 4
1.5.6 Other Hardware Devices 4
1.6 Project Schedule Breakdown 5
CHAPTER TWO: DESIGN AND MODELING 6
2.1 Introduction to Modelling 6
2.2 User Interface Models 6
2.2.1 Landing page 6
2.2.2 Admin Dashboard 6
2.2.3 Receptionist Dashboard 7
2.2.4 Mechanic Dashboard 7

2.2.5 Customer Dashboard 8
2.2.7 Vehicle Registration Form 9
2.2.8 Appointment Form 9
2.2.9 Service Form 10
2.2.10 Inventory Form 10
2.3 Logic Models 11
2.3.1 Login flowchart 11
2.3.2 Use case diagram 12
2.3.3 Entity relationship diagram 12
2.3.4 Dfd level 0 13
2.3.5 Dfd level 1 13
2.3.4 Dfd level 2 14
2.3.7 Vehicle registration dataflow 14
2.3.8 Appointment booking dataflow 15
2.3.9 inventory dataflow 15
2.3.10 Services dataflow 16
CHAPTER THREE: SYSTEM IMPLEMENTATION 17
3.1 Introduction 17
3.2 User Interface Development 17
3.3 Logic Development 17
3.3.1 Main page 17
3.3.2 Admin dashboard 20
3.3.3 Admin inventory 21
3.3.4 Admin job scheduling 23
3.3.5 Admin users management 25
3.3.6 Admin service managment 27
3.3.7 Customer appointments booking 28
3.3.8 Mechanic inventory usage 30
3.3.9 Receptionist vehicle registration 32
3.4 Testing 34
3.4.1 Testing Activities Included: 34
3.5 Deployment 34
CHAPTER FOUR: CONCLUSION AND RECOMMENDATION 36

4.1 Conclusion 36
4.2 Recommendation 36
References 37

CHAPTER ONE: PROJECT PLANNING AND ANALYSIS

1.1 Statement of Problem

Many automotive garages still operate with paper-based processes for scheduling, recordkeeping, and service histories, which creates inefficiencies and delays in daily operations. The use of a traditional paper-based method of running a garage creates inefficiencies and experience delays in their day-to-day operations. The use of paper is not only a labor-intensive process but also leads to errors and a lack of visibility into the process of job workflow management which makes it difficult for garage managers to make timely decisions, and have effective communication on the status of vehicles being repaired, with their staff (Auto Smarts, 2025).

Ineffective recordkeeping and slow service job workflows frustrate customers by creating extended periods of time between when a customer drops off their vehicle and when they receive an indication of what types of service will be performed and when they can expect to pick up their vehicle which ultimately leads to them losing trust in you and possibly seeking out another garage to use as their place of business (Auto Training Centre, 2026).

Garage owners and managers are also having difficulties obtaining accurate productions and operational reporting because they are using a manual paper system, which could result in cash flow losses, improperly tracking of the parts needed to complete jobs, and, ultimately, make poor management decisions (Business Research Insights, 2026). Inefficient processes reduce the productivity of employees, as well as resulting in being less satisfied with a repair experience for the customer; therefore, implementing digital solutions and automated workflow processes are needed for modern day garage operations (Auto Smarts, 2025; Business Research Insights, 2026).

1.2 Study Justification

By developing a Garage Management System (GMS), garages can better manage their operations and benefit from greater efficiency (Deloitte, 2024; McKinsey & Company, 2023). GMS provides digital customer and vehicle records; customers can manage and schedule their own services online; GMS can track all jobs automatically, and it has inventories where the garage tracks everything, all of which

eliminate lost records and delays in providing service due to the lack of communication between customer service and technicians (PwC, 2023). In addition, GMS allows for instantaneous updates about the current status of a service and an inventory and provides technical support to staff (Deloitte, 2024). GMS also digitally tracks customers' service histories, which helps with financial reporting (McKinsey & Company, 2023). Thus, the GMS developed in this study will help reduce problems with inefficiencies previously identified in the study and improve customer satisfaction through improved operational performance and efficiency (PwC, 2023).

1.3 System Objectives

1.3.1 General Objective

(i) To design and develop a Garage Management System.

1.3.2 Specific Objectives

(i) To implement a feature for recording and managing customer and vehicle service details.

(ii) To implement service scheduling and job tracking.

(iii) To implement inventory management for spare parts.

(iv) To implement billing and report generation.

1.4 Functional Requirements

User	**User Activities**	**Features**
Admin	- Staff/ Users management	- Dashboard to register new mechanics and Receptionist
	- Job scheduling	- Dashboard to manage job schedules
	- Report generation	- Database Querry to provide access details
	- Manage services and pricing	- Service and pricing management module
Mechanic	- View assigned jobs	- Dashboard to display assigned repair jobs

	- Update job status	- Job status update interface (e.g., pending, in progress, completed)
	- Add repair notes	- Detail short repair notes
Receptionist	- Register customer's vehicle	- Vehicle registration module with customer details
	- Book appointments for clients	- Appointment Booking system
	- Manage payments	- Payment processing module
Customer	- Book appointments	- Online service booking interface
	- View status	- Real-time job status tracking feature
	- Give feedback	- Customer feedback
	- View history	- Service history and module

Table 1.4 Functional Requirements

1.5 Breakdown of Tools & Resources to Be Used

1.5.1 Laptop

The laptop will be used for system development, testing, and documentation.

Minimum Specifications:

- Processor: Intel Core i5 (8th Generation)
- RAM: 8 GB
- Storage: 256 GB SSD
- Operating System: Windows 11 or Linux
- Display: 14" or higher

1.5.2 Front End

The front end will be used to design and develop the user interface of the system.

Languages:

- HTML – for page structure
- CSS – for styling and layout
- JavaScript – for interactivity and validation

1.5.3 IDE (Integrated Development Environment)

The IDE will be used to write, debug, and manage the source code efficiently.

IDE to be Used:

- Visual Studio

1.5.4 Back End

The back end will handle business logic, database operations, and user authentication.

Technologies:

- Programming Language: PHP
- Database: MySQL
- Server: Apache (XAMPP)

1.5.5 Internet

Internet connectivity will be required for development, testing, updates, and deployment.

Specifications:

- Minimum Speed: 10 Mbps
- Recommended Speed: 20 Mbps or higher
- Connection Type: Broadband or Fiber

1.5.6 Other Hardware Devices

External Storage (Flash drive or external HDD) – for backups

1.6 Project Schedule Breakdown

	PROJECT MILESTONES					
WEEKS	**Project Planning & Analysis** **(System Documentation: Cover page & Chapter One)**	**Project Design & Modeling** **(System Documentation Chapter Two)**	**Project Development & Testing** **(System Documentation Chapter Three)**	**Project Deployment** **(System Documentation Chapter Three)**	**Final Touches of System Documentation** **(Preliminary Pages, Chapter Four & References)**	**Project Presentation**
7-9 Jan						
12-16Jan						
19-23 Jan						
26-30 Jan						
2-6 Feb						
9-13 Feb						
16-20 Feb						
23-27 Feb						
2-6 Mar						
9-13 Mar						
16-20Mar						
23-27Mar						
3rd April						

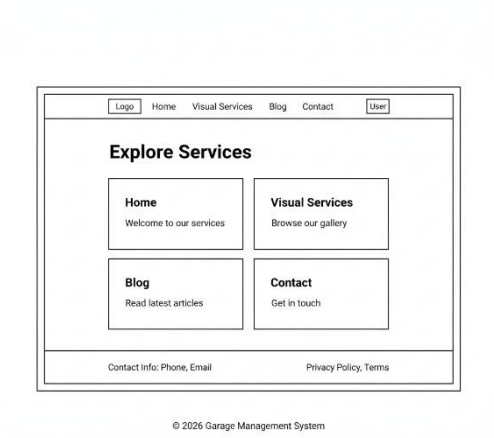
Table 2.6 Project Schedule Breakdown

CHAPTER TWO: DESIGN AND MODELING

2.1 Introduction to Modelling

In the chapter, design and model creation was used to illustrate a Garage Management System design before the system is built. Modeling offered the ability to visualize the overall system, understand the major components in the system, and describe interaction between different components in the system. Building models prior to building the Garage Management System made it easier to make decisions about how to build the system, identify potential problems before they occur, and provided for a more organized and efficient development process.

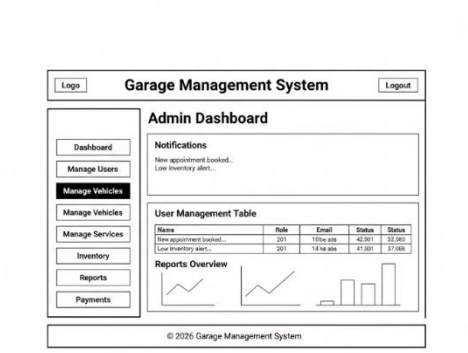
2.2 User Interface Models



This is the main page for all users.it contains basic details about the garage.

Figure 2.2. 1 Main Page

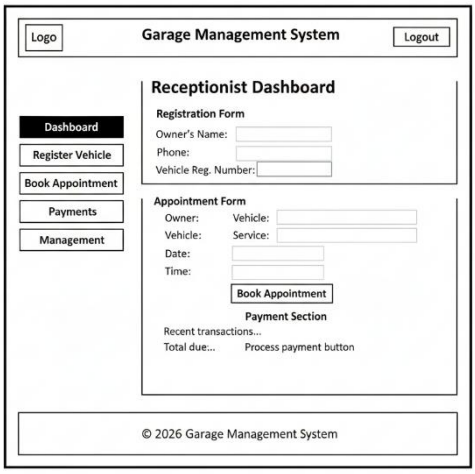
2.2.2 Admin Dashboard



This is the admin panel. Admins can register staff (receptionist and mechanic) schedule jobs to mechanic, manage inventory and services and also view reports.

Figure 2.2. 2 Admin Dashboard

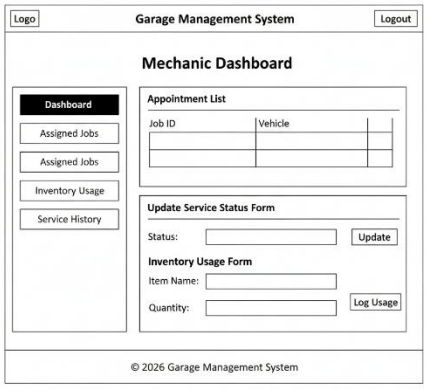
2.2.3 Receptionist Dashboard



This is the receptionist panel. Receptionist can register vehicles, schedule appointments, manage payments, view vehicle logs and give feedback to customers.

Figure 2.2. 3 Receptionist Dashboard

2.2.4 Mechanic Dashboard



This is mechanic panel. Mechanics are able to view jobs assigned,give feedback and recommendations of services performed.

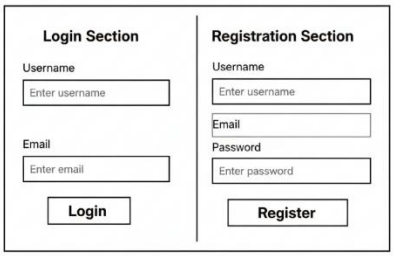
Figure 2.2. 4 Mechanic Dashboard

2.2.5 Customer Dashboard

This is the customers panel. customer are able to book services, view status, history and also give feedback.

Figure 2.2. 5 Customer Dashboard

2.2.6 Login form



This shows both login and registration form design using email,username and password as input details**.**

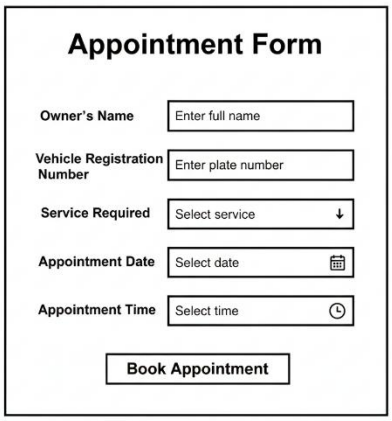
Figure 2.2. 6 Login/Registration form

2.2.7 Vehicle Registration Form

Captures owner details, validates formats, saves to database, and confirms successful registration

Figure 2.2. 7 Vehicle Registration Form

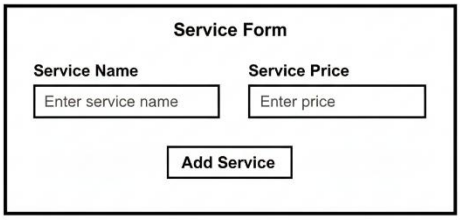
2.2.8 Appointment Form



Guides booking process with input fields, validation, database save, and confirmation message

Figure 2.2. 8 Appointment Registration Form

2.2.9 Service Form



Simple forms for adding services or inventory items, validated and stored in respective databases.

Figure 2.2. 9 Service Form

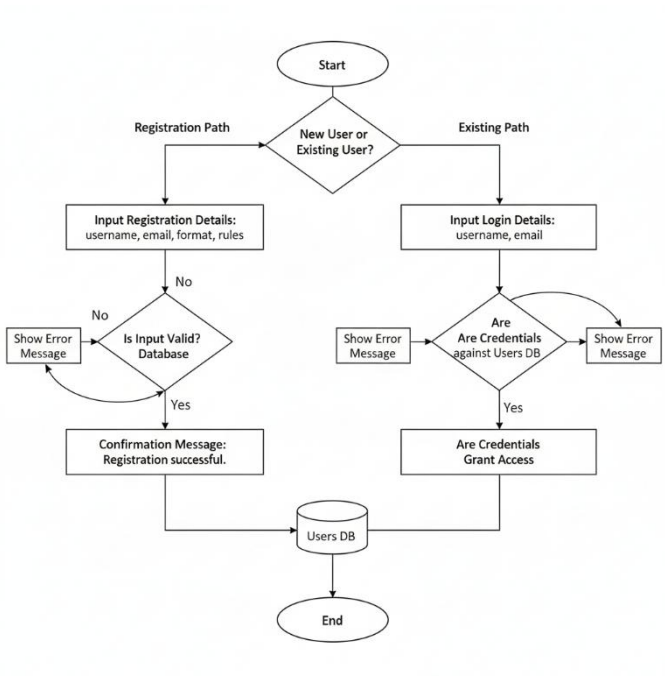
2.2.10 Inventory Form

Shows input of item details, validation of name and price, saving to the inventory database, and confirmation message

Figure 2.2. 10 Inventory Form

2.3 Logic Models

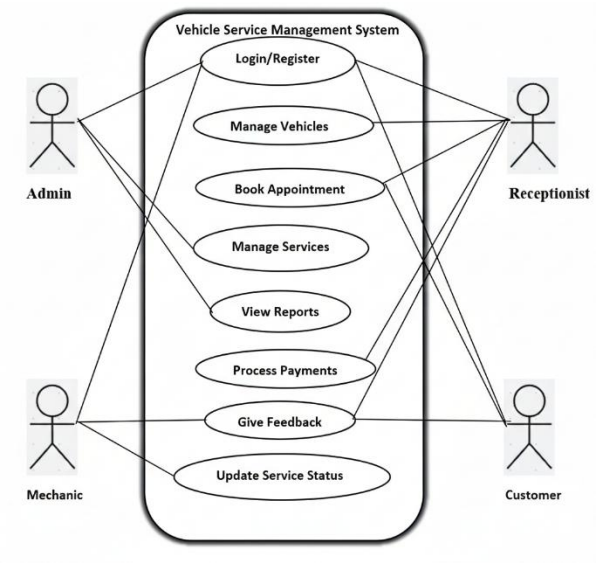
2.3.1 Login flowchart



Captures user credentials, validates against stored records, and either grants access or returns an error message.

Figure 2.3. 1 Login Flowchart

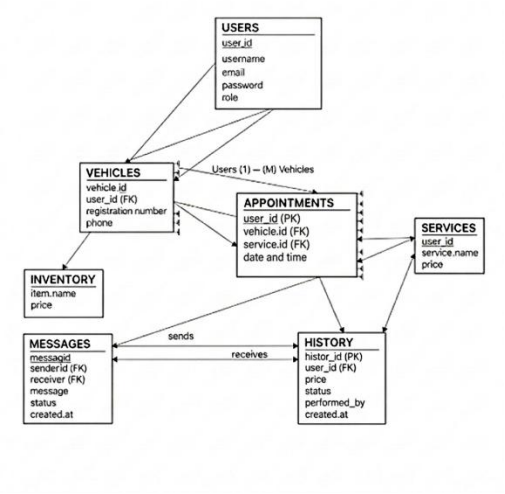
2.3.2 Use case diagram



Depicts Admin, Receptionist, Mechanic, and Customer actors connected to system functions like login, vehicle management, appointments, services, payments, reports, feedback, and service updates.

Figure 2.3. 2 Use Case Diagram

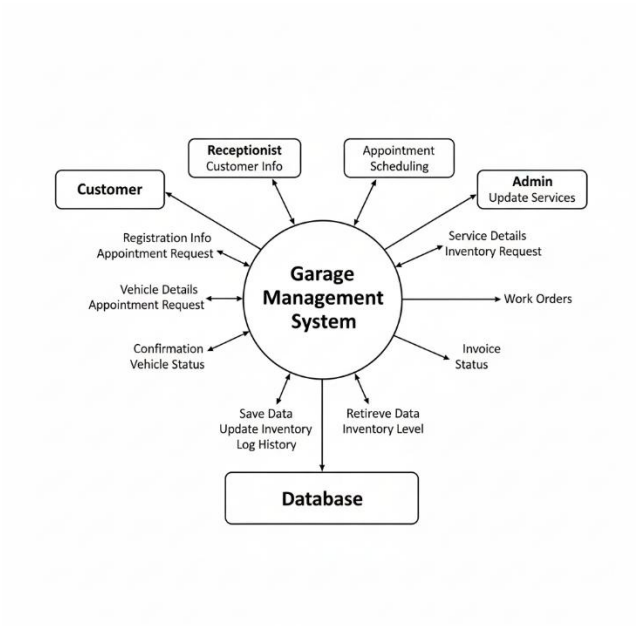
2.3.3 Entity relationship diagram



Connects customers, vehicles, and services to scheduled appointments in a structured schema.

Figure 2.3. 3 Entity Relationship Diagram

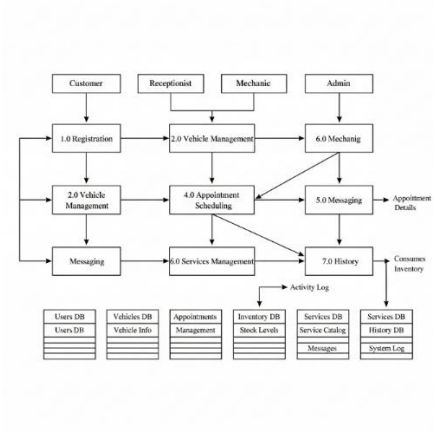
2.3.4 Dfd level 0



High-level overview showing data flow between users (Admin, Receptionist, Mechanic, Customer) and the system.

Figure 2.3. 4 Dfd Level 0

2.3.5 Dfd level 1



Breaks down processes like registration, booking, payments, and service updates with data stores.

Figure 2.3. 5 Dfd Level 1

2.3.4 Dfd level 2

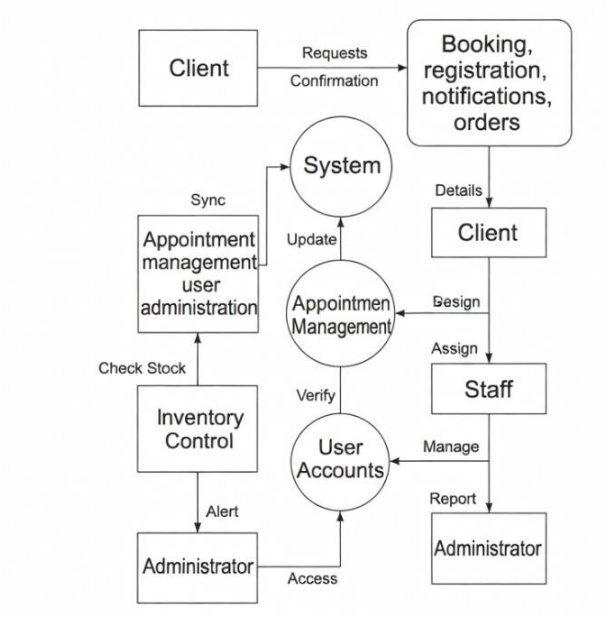
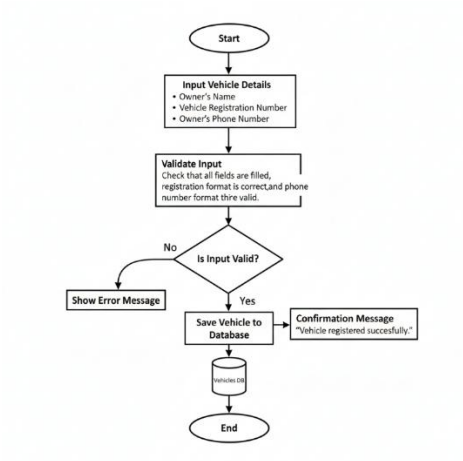


Figure 2.3. 6 Dfd Level 2

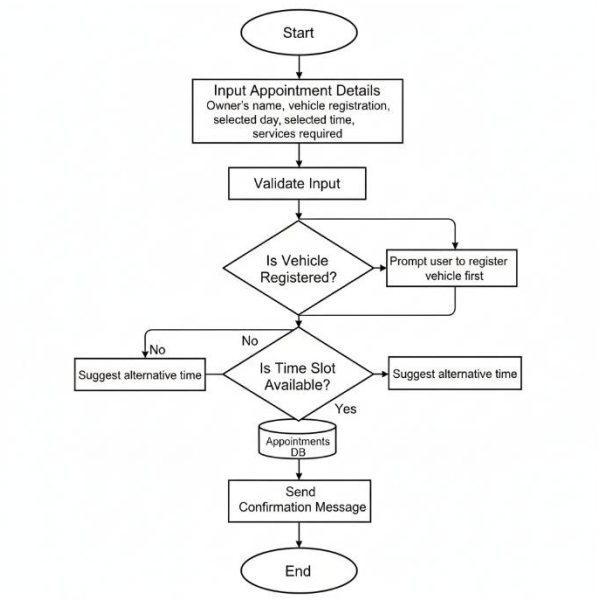
2.3.7 Vehicle registration dataflow



Captures owner details, validates formats, saves to database, and confirms successful registration

Figure 2.3. 7 vehicle Registration Dataflow

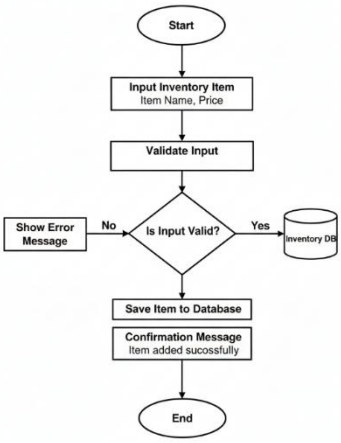
2.3.8 Appointment booking dataflow



Guides booking process with input fields, validation, database save, and confirmation message

Figure 2.3. 8 Appointment Booking Dataflow

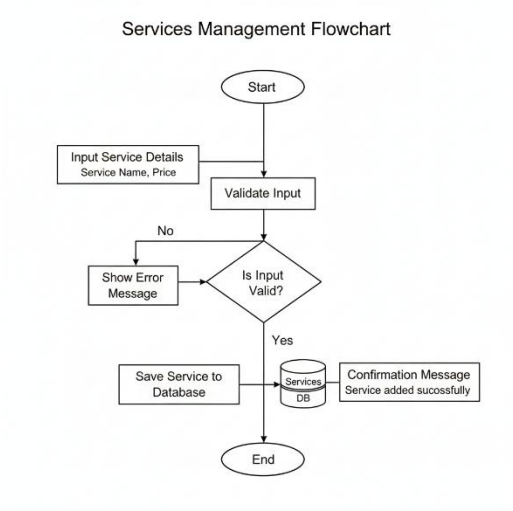
2.3.9 inventory dataflow



Shows input of item details, validation of name and price, saving to the inventory database, and confirmation message

Figure 2.3. 9 Inventory Dataflow

2.3.10 Services dataflow



This shows how data flow for service form for adding services or inventory items, validated and stored in respective databases.

Figure 2.3. 10 Service Dataflow

CHAPTER THREE: SYSTEM IMPLEMENTATION

3.1 Introduction

This chapter explains the process followed during the development, testing, and deployment of the Garage Management System. It covers the creation of the user interface, the system logic that processes data, the testing of features, and how the system was made available online.

3.2 User Interface Development

The user interface is the part of the system clients interact with. The system was designed to be elegant, welcoming, and easy to navigate. The UI includes appointment booking and registration pages, service listing pages and a management dashboard for the Garage team. HTML and CSS ensure a smooth, modern, and visually appealing experience that reflects the auto garage brand.

3.3 Logic Development

3.3.1 Main page

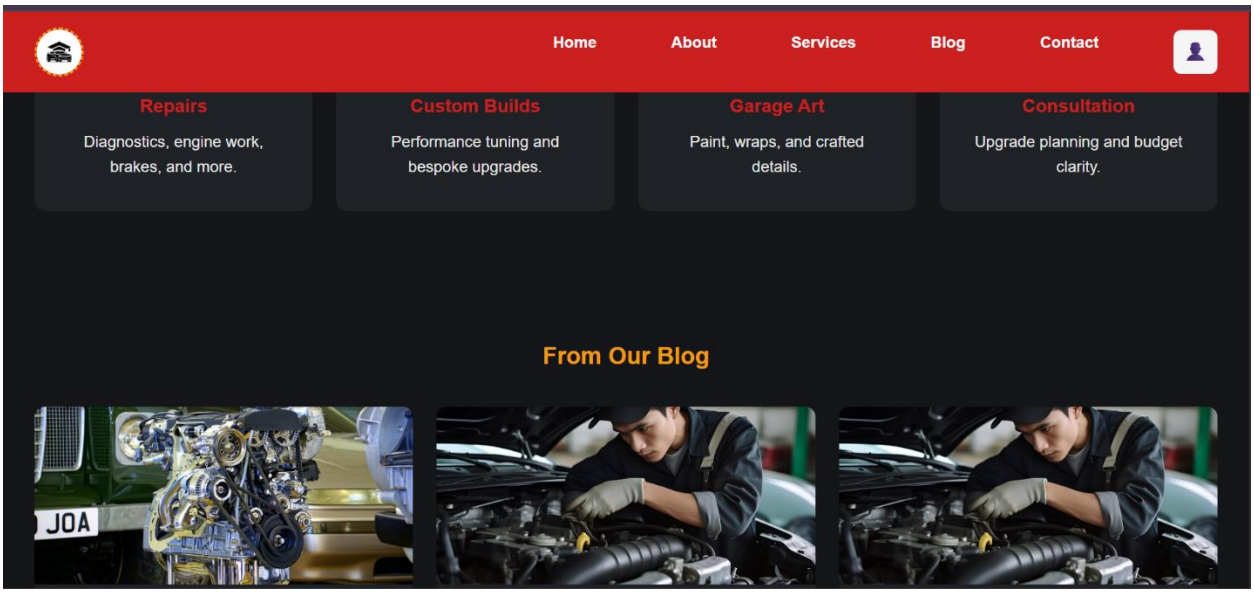


Figure 3.3. 1 Main Page

Main page source code:

```html
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8" />
  <meta name="viewport" content="width=device-width, initial-scale=1.0" />
  <link rel="stylesheet" href="css/main.css">
  <link rel="stylesheet" href="https://cdnjs.cloudflare.com/ajax/libs/font-awesome/6.5.0/css/all.min.css">
  <title>Alpha Garage</title>
</head>
<body>

  <!-- Header -->
  <header>
    <nav>
      <!-- Logo -->
      <div class="logo">
        <img src="images/l1 (1).jpeg" alt="Alpha Garage Logo">
      </div>

      <!-- Navigation -->
      <ul id="navLinks">
        <li><a href="#home">Home</a></li>
        <li><a href="#about">About</a></li>
        <li><a href="#services">Services</a></li>
        <li><a href="#blog">Blog</a></li>
        <li><a href="#contact">Contact</a></li>

        <!-- Add User Icon -->
        <li class="mobile-user-icon">
```

```html
<body>
  <header>
    <nav>
      <ul id="navLinks">
        <li class="mobile-user-icon">
        </li>
      </ul>

      <!-- Mobile Menu Icon -->
      <div class="menu-icon" id="menuIcon">&#9776;</div>
    </nav>
  </header>

  <!-- Hero Section -->
  <section id="home">
    <div class="hero-background">
      <img src="images/img 1.jpg" alt="Garage Hero Background">
    </div>
    <div class="hero-content">
      <h1>ALPHA GARAGE</h1>
      <h1>Crafting Garage Art & Services</h1>
      <p>Where mechanics meet creativity</p>
      <a href="#services">Explore Services</a>
    </div>
  </section>

  <!-- About Section -->
  <section id="about">
    <div class="about-wrapper">
```

```html
<body>
  <section id="blog">
    </div>
  </section>

  <!-- Contact Section (Updated to Social Media Handles) -->
  <section id="contact">
    <div class="social-wrapper">
      <h2>Remember  To Follow All Our Social Handles</h2>
      <div class="social-links">
        <a href="https://www.tiktok.com/@alphagarage" target="_blank"><i class="fab fa-tiktok"></i> TikTok</a>
        <a href="https://www.instagram.com/alphagarage" target="_blank"><i class="fab fa-instagram"></i> Instagram</a
        <a href="https://www.facebook.com/alphagarage" target="_blank"><i class="fab fa-facebook"></i> Facebook</a>
        <a href="https://wa.me/254700000000" target="_blank"><i class="fab fa-whatsapp"></i> WhatsApp</a>
        <h2>Connect With Us</h2>
      </div>
    </div>

    <!-- Image beside socials -->
    <div class="map-wrapper">
      <div class="map-header">
        <h3>Our Location</h3>
        <p>Visit us at the garage</p>
      </div>
      <div class="map-placeholder">
        <img src="Images/map.jpeg" alt="Map Placeholder">
      </div>
    </div>
  </section>
```

```html
<body>
  <section id="contact">
  </section>

  <!-- Footer -->
  <footer>
    <div class="social-icons">
      <a href="#"><i class="fab fa-facebook"></i></a>
      <a href="#"><i class="fab fa-twitter"></i></a>
      <a href="#"><i class="fab fa-instagram"></i></a>
      <a href="#"><i class="fab fa-linkedin"></i></a>
    </div>
    <p>© 2026 Alpha Garage. All rights reserved.</p>
  </footer>

  <!-- Mobile Menu Script -->
  <script>
    const menuIcon = document.getElementById("menuIcon");
    const navLinks = document.getElementById("navLinks");

    menuIcon.addEventListener("click", () => {
      navLinks.classList.toggle("show");
    });
  </script>

</body>
</html>
```

Figure 3.3. 2 Main Page source code

3.3.2 Admin dashboard

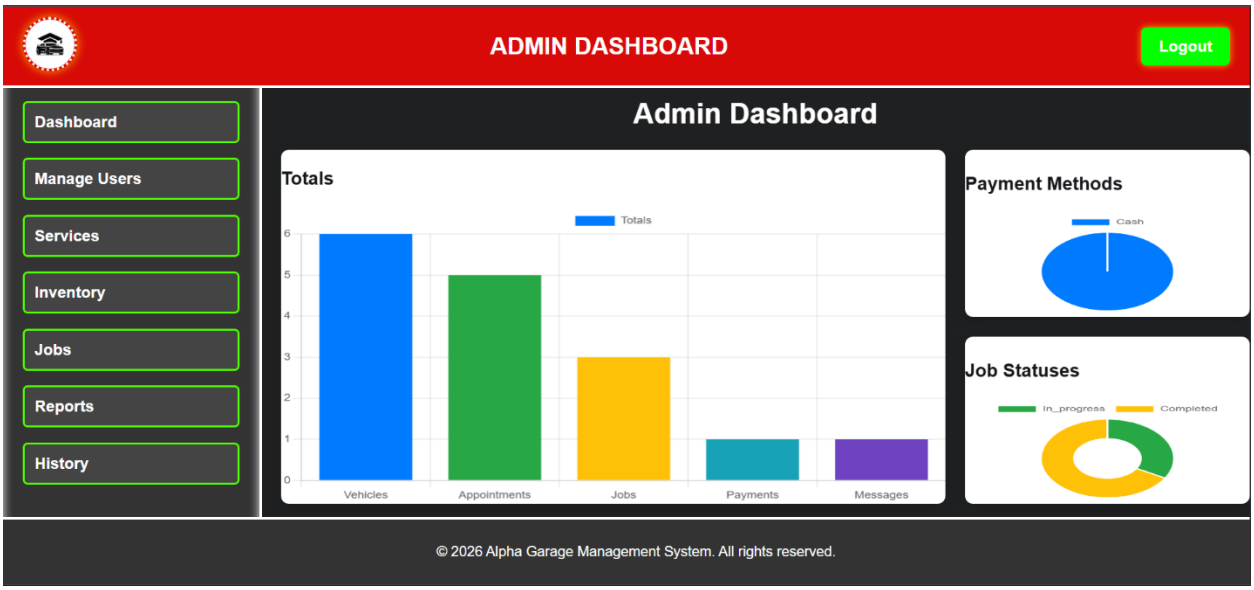


Figure 3.3. 3 Admin Dashboard

Admin dashboard source Code:

```
<main class="main-content">
    <h2 class="page-heading"><?= $page_title; ?></h2>

    <!-- Flex wrapper -->
    <div class="dashboard-flex">
        <!-- Left side: Totals -->
        <div class="totals-box chart-container">
            <h3>Totals</h3>
            <canvas id="totalsChart"></canvas>
        </div>

        <!-- Right side: stacked column -->
        <div class="right-box">
            <div class="chart-container">
                <h3>Payment Methods</h3>
                <canvas id="paymentChart"></canvas>
            </div>
            <div class="chart-container">
                <h3>Job Statuses</h3>
                <canvas id="jobChart"></canvas>
            </div>
        </div>
    </div>

    <!-- Appointments Line Chart -->
    <div class="chart-container" style="max-width:800px; margin-top:30px;">
        <h3>Appointments (Last 7 Days)</h3>
        <canvas id="appointmentsChart"></canvas>
```

```
    <!-- Appointments Line Chart -->
    <div class="chart-container" style="max-width:800px; margin-top:30px;">
        <h3>Appointments (Last 7 Days)</h3>
        <canvas id="appointmentsChart"></canvas>
    </div>

    <!-- Inventory Bar Chart -->
    <div class="chart-container" style="max-width:800px; margin-top:30px;">
        <h3>Inventory Levels</h3>
        <canvas id="inventoryChart"></canvas>
    </div>
/main>

script src="https://cdn.jsdelivr.net/npm/chart.js"></script>
script>
/ Totals Bar Chart
ew Chart(document.getElementById('totalsChart'), {
    type: 'bar',
    data: {
        labels: ['Vehicles','Appointments','Jobs','Payments','Messages'],
        datasets: [{
            label: 'Totals',
            data: [<?= $totalVehicles; ?>, <?= $totalAppointments; ?>, <?= $totalJobs; ?>, <?= $totalPayments; ?>, <?
            backgroundColor: ['#007bff','#28a745','#ffc107','#17a2b8','#6f42c1']
        }]
    },
    options: { responsive: true, maintainAspectRatio: false }
```

Figure 3.3. 4 Admin Dashboard Source Code

3.3.3 Admin inventory

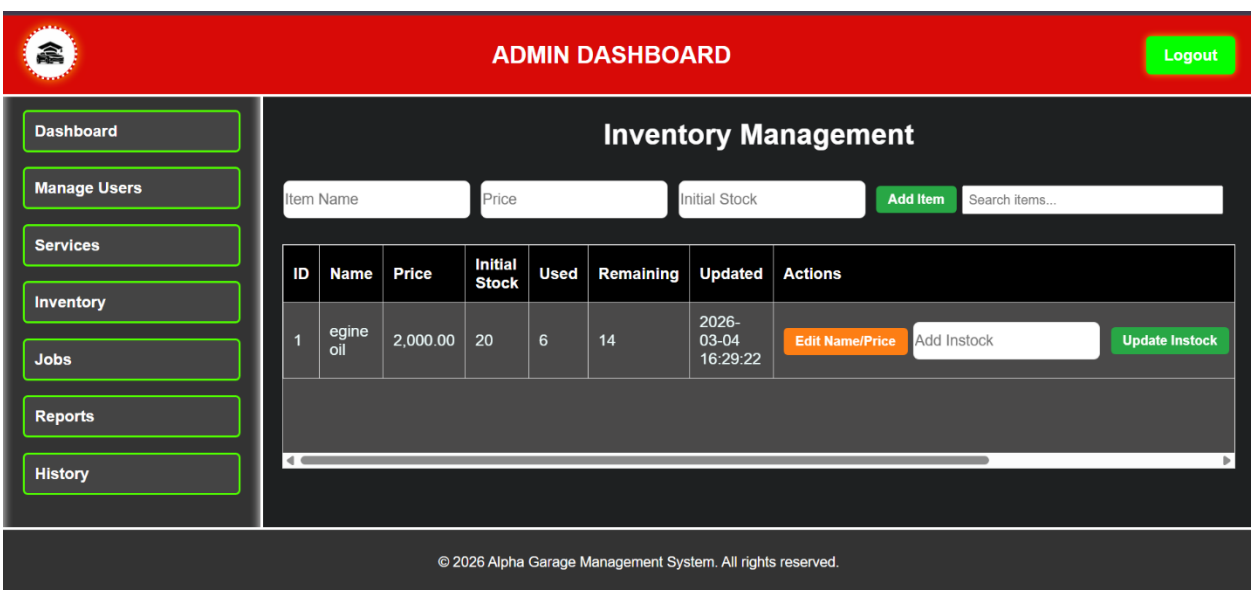


Figure 3.3. 5 Admin Inventory

Admin inventory source Code:

```
<main class="main-content">
    <h2 class="page-heading">
<?= $page_title; ?></h2>

    <!-- Popup Messages -->
    <?php include_once("../public/includes/popup.php"); ?>

    <!-- Add Item Form -->
    <form method="POST" class="inline-form">
        <input type="text" name="name" placeholder="Item Name" required>
        <input type="number" step="0.01" name="price" placeholder="Price" required>
        <input type="number" name="stock" placeholder="Initial Stock" required>
        <button type="submit" name="add_item" class="btn-green">Add Item</button>
    </form>

    <!-- Search Box (always stays here) -->
    <input type="text" id="searchBox" placeholder="Search items..." style="margin:15px 0; padding:5px; width:250px;">

    <!-- Update Item Form (appears below search bar when triggered) -->
    <form method="POST" class="inline-form" id="updateForm" style="margin-top:15px; display:none;">
        <input type="hidden" name="item_id" id="update_item_id">
        <input type="text" name="name" id="update_name" placeholder="Item Name" required>
        <input type="number" step="0.01" name="price" id="update_price" placeholder="Price" required>
        <button type="submit" name="update_item" class="btn-blue">Update Name/Price</button>
    </form>

    <!-- Inventory Table -->
    <div class="table-wrapper">
        <div class="table-container">
```

```
<main class="main-content">
    <!-- Inventory Table -->
    <div class="table-wrapper">
        <div class="table-container">
            <table class="styled-table" id="inventoryTable">
                <thead>
                    <tr>
                        <th>ID</th><th>Name</th><th>Price</th><th>Initial Stock</th>
                        <th>Used</th><th>Remaining</th><th>Updated</th><th>Actions</th>
                    </tr>
                </thead>
                <tbody>
                    <?php while($row = $inventory->fetch_assoc()): ?>
                    <tr>
                        <td><?= $row['id']; ?></td>
                        <td><?= htmlspecialchars($row['item_name']); ?></td>
                        <td><?= number_format($row['price'],2); ?></td>
                        <td><?= $row['instock']; ?></td>
                        <td><?= $row['used']; ?></td>
                        <td><?= $row['remaining']; ?></td>
                        <td><?= $row['updated_at']; ?></td>
                        <td>
                            <!-- Select Item for Update -->
                            <button type="button" class="btn-orange"
                                onclick="selectItem(<?= $row['id']; ?>, '<?= htmlspecialchars($row['item_name']); ?>'
                                Edit Name/Price
                            </button>
                            <!-- Update Instock -->
                            <form method="POST" class="inline-form">
```

```
<main class="main-content">
    <div class="table-wrapper">
        <div class="table-container">
            <table class="styled-table" id="inventoryTable">
                <tbody>
                                onclick="selectItem(<?= $row['id']; ?>, '<?= htmlspecialchars($row['item_name']); ?>'
                                Edit Name/Price
                            </button>
                            <!-- Update Instock -->
                            <form method="POST" class="inline-form">
                                <input type="hidden" name="item_id" value="<?= $row['id']; ?>">
                                <input type="number" name="new_stock" placeholder="Add Instock" required>
                                <button type="submit" name="update_instock" class="btn-green">Update Instock</button>
                            </form>
                            <!-- Update Used -->
                            <form method="POST" class="inline-form">
                                <input type="hidden" name="item_id" value="<?= $row['id']; ?>">
                                <input type="number" name="add_used" placeholder="Add Used" required>
                                <button type="submit" name="update_used" class="btn-orange">Update Used</button>
                            </form>
                        </td>
                    </tr>
                    <?php endwhile; ?>
                </tbody>
            </table>
        </div>
    </div>
</main>
```

Figure 3.3. 6 Admin Inventory Source Code

3.3.4 Admin job scheduling

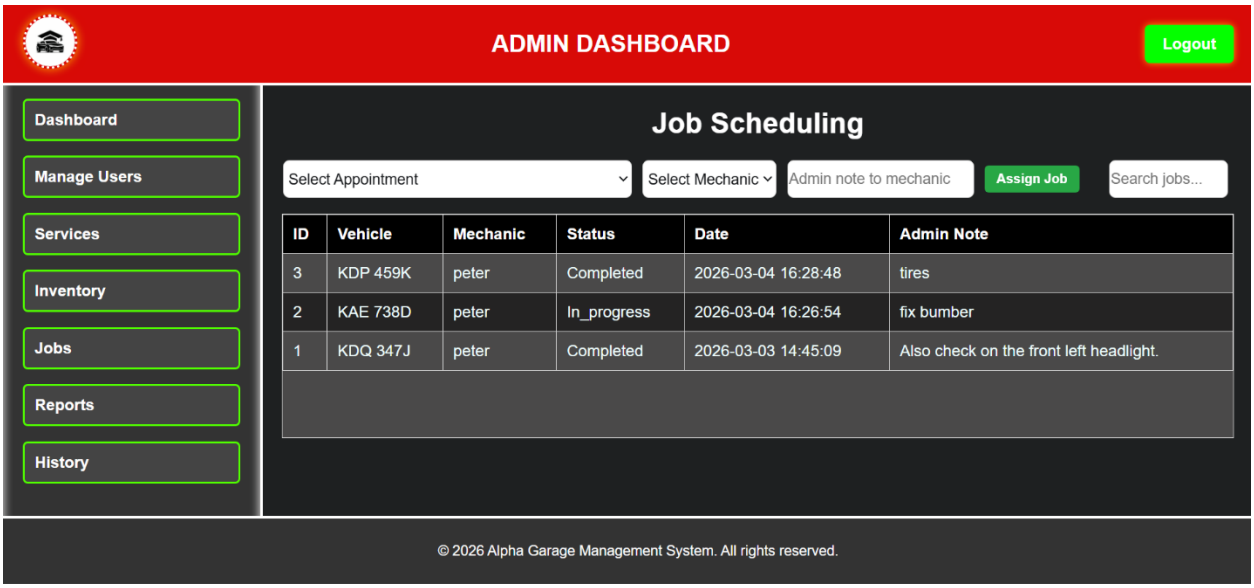


Figure 3.3. 7 Admin job scheduling

Admin job scheduling source Code:

```
<main class="main-content">
    <h2 class="page-heading">
<?= $page_title; ?></h2>

    <!-- Popup Messages -->
    <?php include_once("../public/includes/popup.php"); ?>

    <!-- Assign Job Form -->
    <form method="POST" class="inline-form">
        <select name="appointment_id" required onchange="updateVehicleId(this)">
            <option value="">Select Appointment</option>
            <?php while($a = $appointments->fetch_assoc()): ?>
                <option value="<?= $a['id']; ?>" data-vehicle="<?= $a['vehicle_id']; ?>" data-owner="<?= $a['owner_id
                    <?= $a['reg_number']; ?> (Owner: <?= htmlspecialchars($a['owner']); ?>, <?= $a['appointment_date'
                </option>
            <?php endwhile; ?>
        </select>
        <input type="hidden" name="vehicle_id" id="vehicle_id">
        <input type="hidden" name="owner_id" id="owner_id">

        <select name="mechanic_id" required>
            <option value="">Select Mechanic</option>
            <?php while($m = $mechanics->fetch_assoc()): ?>
                <option value="<?= $m['id']; ?>"><?= htmlspecialchars($m['username']); ?></option>
            <?php endwhile; ?>
        </select>
        <input type="text" name="notes" placeholder="Admin note to mechanic">
        <button type="submit" name="assign_job" class="btn-green">Assign Job</button>
        <!-- Search Box -->
```

```
    <!-- Jobs Table -->
    <div class="table-wrapper">
        <div class="table-container">
            <table class="styled-table" id="jobsTable">
                <thead>
                    <tr>
                        <th>ID</th><th>Vehicle</th><th>Mechanic</th><th>Status</th><th>Date</th><th>Admin Note</th>
                    </tr>
                </thead>
                <tbody>
                    <?php while($row = $jobs->fetch_assoc()): ?>
                    <tr>
                        <td><?= $row['id']; ?></td>
                        <td><?= $row['reg_number']; ?></td>
                        <td><?= htmlspecialchars($row['mechanic']); ?></td>
                        <td><?= ucfirst($row['status']); ?></td>
                        <td><?= $row['created_at']; ?></td>
                        <td><?= htmlspecialchars($row['notes']); ?></td>
                    </tr>
                    <?php endwhile; ?>
                </tbody>
            </table>
        </div>
    </div>
</main>
```

Figure 3.3. 8 Admin Job Scheduling Source Code

3.3.5 Admin users management

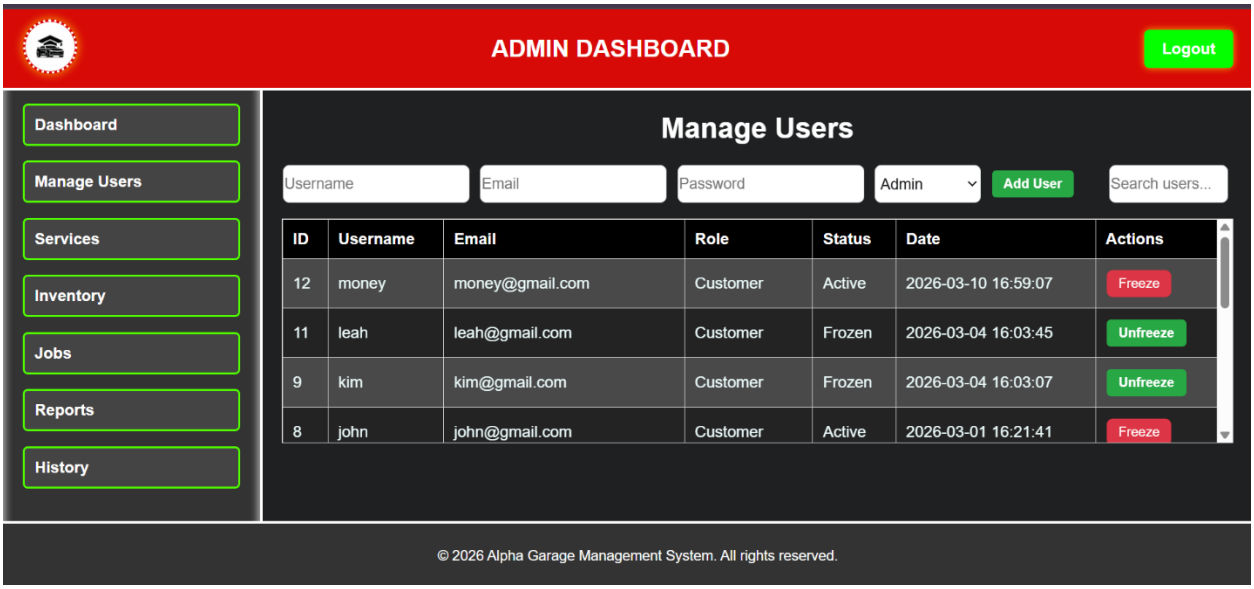


Figure 3.3. 9 Admin users management

Admin user management source code:

```
<main class="main-content">
    <h2 class="page-heading">
<?= $page_title; ?>
</h2>

    <?php include_once("../public/includes/popup.php"); ?>

    <!-- Add User Form -->
    <form method="POST" class="inline-form">
        <input type="text" name="username" placeholder="Username" required>
        <input type="email" name="email" placeholder="Email" required>
        <input type="password" name="password" placeholder="Password" required>
        <select name="role" required>
            <option value="admin">Admin</option>
            <option value="receptionist">Receptionist</option>
            <option value="mechanic">Mechanic</option>
            <option value="customer">Customer</option>
        </select>
        <button type="submit" name="add_user" class="btn-green">Add User</button>
    </form>

    <!-- Search Box -->
    <input type="text" id="searchUsers" placeholder="Search users..." style="margin-left:30px; width:120px">

    <!-- Users Table -->
    <div class="table-wrapper">
        <div class="table-container">
            <table class="styled-table" id="usersTable">
                <thead>
                    <tr>
```

```
<main class="main-content">
    <div class="table-wrapper">
        <div class="table-container">
            <table class="styled-table" id="usersTable">
                <tbody>
                    <?php while($row = $users->fetch_assoc()): ?>
                    <tr>
                        <td><?= $row['id']; ?></td>
                        <td><?= htmlspecialchars($row['username']); ?></td>
                        <td><?= htmlspecialchars($row['email']); ?></td>
                        <td><?= ucfirst($row['role']); ?></td>
                        <td><?= ucfirst($row['status']); ?></td>
                        <td><?= $row['created_at']; ?></td>
                        <td>
                            <form method="POST" class="inline-form">
                                <input type="hidden" name="user_id" value="<?= $row['id']; ?>">
                                <input type="hidden" name="current_status" value="<?= $row['status']; ?>">
                                <?php if ($row['status'] === 'active'): ?>
                                    <button type="submit" name="toggle_status" class="btn-red">Freeze</button>
                                <?php else: ?>
                                    <button type="submit" name="toggle_status" class="btn-green">Unfreeze</button>
                                <?php endif; ?>
                            </form>
                        </td>
                    </tr>
                    <?php endwhile; ?>
                </tbody>
            </table>
```

```
<main class="main-content">
    <div class="table-wrapper">
        <div class="table-container">
            <table class="styled-table" id="usersTable">
                <tbody>
                                <input type="hidden" name="current_status" value="<?= $row['status']; ?>">
                                <?php if ($row['status'] === 'active'): ?>
                                    <button type="submit" name="toggle_status" class="btn-red">Freeze</button>
                                <?php else: ?>
                                    <button type="submit" name="toggle_status" class="btn-green">Unfreeze</button>
                                <?php endif; ?>
                            </form>
                        </td>
                    </tr>
                    <?php endwhile; ?>
                </tbody>
            </table>
        </div>
    </div>
</main>

<script>
// Search functionality for Users
document.getElementById('searchUsers').addEventListener('keyup', function() {
    let filter = this.value.toLowerCase();
    let rows = document.querySelectorAll('#usersTable tbody tr');
    rows.forEach(row => {
        let text = row.textContent.toLowerCase();
```

Figure 3.3. 10 Admin Users Management Source Code

3.3.6 Admin service managment

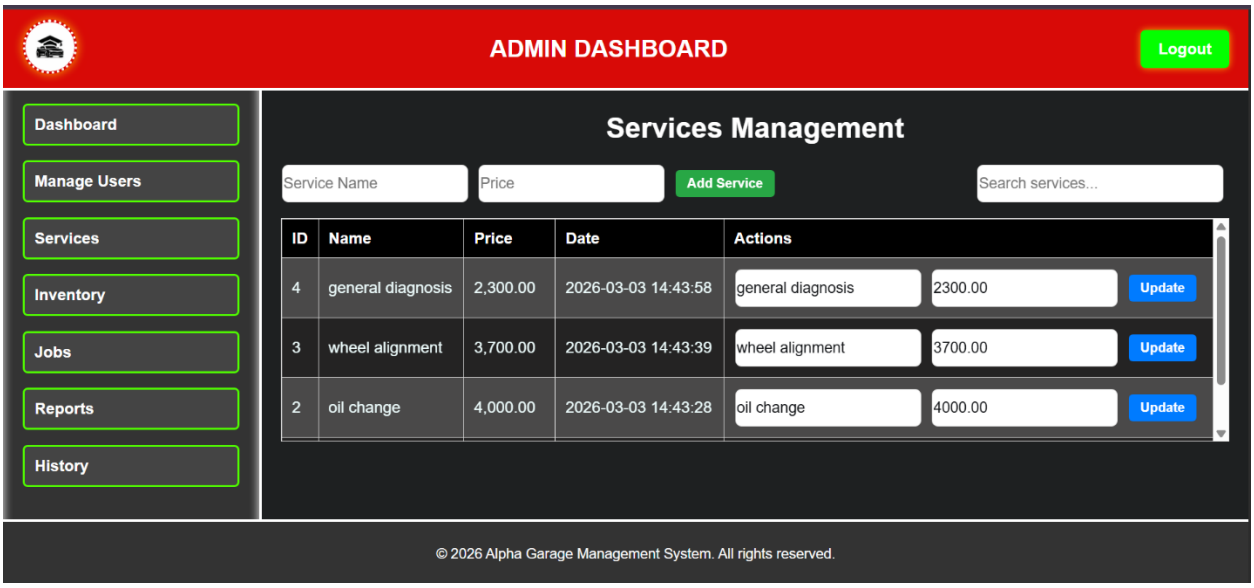


Figure 3.3. 11 Admin service management

Admin service management source Code:

```
<main class="main-content">
    <h2 class="page-heading"><?= $page_title; ?></h2>

    <?php include_once("../public/includes/popup.php"); ?>

    <!-- Add Service Form -->
    <form method="POST" class="inline-form">
        <input type="text" name="service_name" placeholder="Service Name" required>
        <input type="number" step="0.01" name="price" placeholder="Price" required>
        <button type="submit" name="add_service" class="btn-green">Add Service</button>
    </form>

    <!-- Search Box -->
    <input type="text" id="searchServices" placeholder="Search services..." style="margin-left:200px; width:250px;">

    <!-- Services Table -->
    <div class="table-wrapper">
        <div class="table-container">
            <table class="styled-table" id="servicesTable">
                <thead>
                    <tr><th>ID</th><th>Name</th><th>Price</th><th>Date</th><th>Actions</th></tr>
                </thead>
                <tbody>
                    <?php while($row = $services->fetch_assoc()): ?>
                    <tr>
                        <td><?= $row['id']; ?></td>
                        <td><?= htmlspecialchars($row['service_name']); ?></td>
                        <td><?= number_format($row['price'],2); ?></td>
                        <td><?= $row['created_at']; ?></td>
```

```php
<main class="main-content">
    <div class="table-wrapper">
        <div class="table-container">
            <table class="styled-table" id="servicesTable">
                <tbody>
                    <tr>
                        <td><?= $row['id']; ?></td>
                        <td><?= htmlspecialchars($row['service_name']); ?></td>
                        <td><?= number_format($row['price'],2); ?></td>
                        <td><?= $row['created_at']; ?></td>
                        <td>
                            <form method="POST" class="inline-form">
                                <input type="hidden" name="service_id" value="<?= $row['id']; ?>">
                                <input type="text" name="service_name" value="<?= htmlspecialchars($row['service_name
                                <input type="number" step="0.01" name="price" value="<?= $row['price']; ?>" required>
                                <button type="submit" name="update_service" class="btn-blue">Update</button>
                            </form>
                        </td>
                    </tr>
                    <?php endwhile; ?>
                </tbody>
            </table>
        </div>
    </div>
</main>
```

Figure 3.3. 12 Admin Service Management Source Code

3.3.7 Customer appointments booking

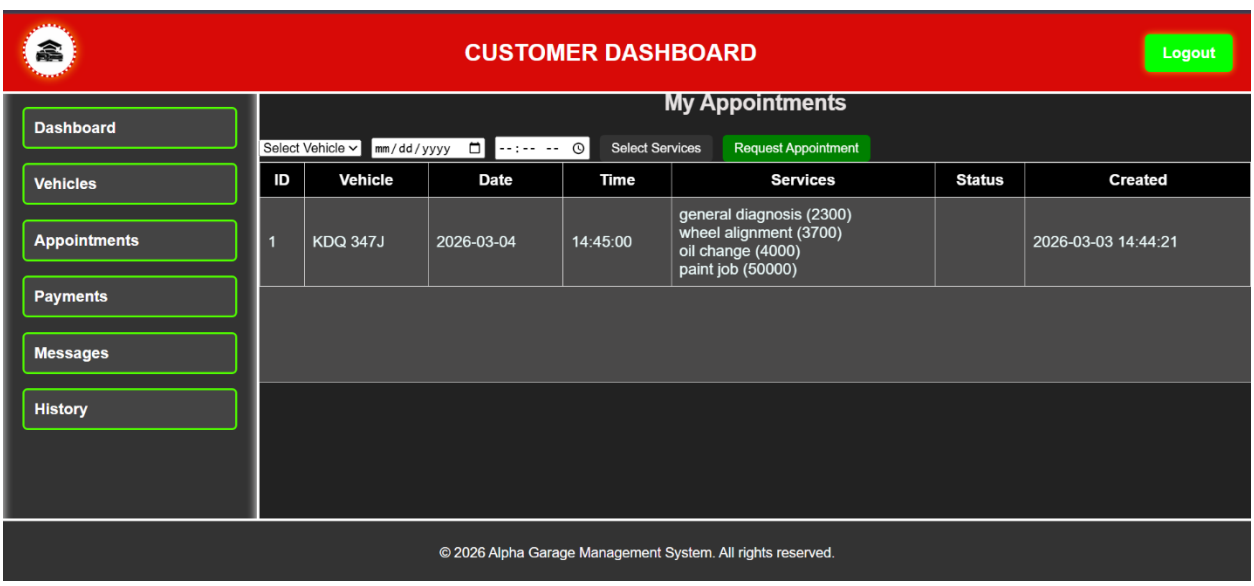


Figure 3.3. 13 Customer appointment booking

Customer appointment booking source Code:

```
<main class="main-content">
    <h2 class="page-heading"><?= $page_title; ?></h2>

    <!-- Popup Messages -->
    <?php include_once("../public/includes/popup.php"); ?>

    <!-- Schedule Appointment Form -->
    <form method="POST" class="inline-form" style="display:flex; gap:10px; align-items:center; flex-wrap:wrap;">
        <!-- Vehicle Selection (searchable) -->
        <input list="vehicleList" name="vehicle_id" placeholder="Search Vehicle (Reg + Owner)" required>
        <datalist id="vehicleList">
            <?php while($v = $vehicles->fetch_assoc()): ?>
                <option value="<?= $v['id']; ?>"><?= $v['reg_number']; ?> (Owner: <?= htmlspecialchars($v['owner']);
            <?php endwhile; ?>
        </datalist>

        <!-- Appointment Date -->
        <input type="date" name="appointment_date" min="<?= date('Y-m-d'); ?>" required>

        <!-- Appointment Time -->
        <input type="time" name="appointment_time" required>

        <!-- Services Dropdown with Search + Checkboxes -->
        <div class="dropdown">
            <button type="button" class="btn-blue dropdown-toggle" onclick="toggleServices()">Select Services</button
            <div id="servicesDropdown" class="dropdown-menu" style="display:none; border:1px solid #ccc; padding:10
                <input type="text" id="serviceSearch" placeholder="Search services..." onkeyup="filterServices()" sty
                <div class="services-grid" id="servicesList">
                    <?php while($s = $services->fetch_assoc(): ?>
```

```
<main class="main-content">
    <form method="POST" class="inline-form" style="display:flex; gap:10px; align-items:center; flex-wrap:wrap;">
        <button type="submit" name="schedule_appointment" class="btn-green">Schedule</button>

        <!-- Inline Search beside Schedule button -->
        <input type="text" id="searchAppointments" placeholder="Search appointments...">
    </form>

    <!-- Appointments Table -->
    <div class="table-wrapper">
        <div class="table-container">
            <table class="styled-table" id="appointmentsTable">
                <thead><tr><th>ID</th><th>Vehicle</th><th>Date</th><th>Time</th><th>Services</th><th>Status</th><th>C
                <tbody>
                    <?php while($row = $appointments->fetch_assoc(): ?>
                    <tr>
                        <td><?= $row['id']; ?></td>
                        <td><?= $row['reg_number']; ?></td>
                        <td><?= $row['appointment_date']; ?></td>
                        <td><?= $row['appointment_time']; ?></td>
                        <td>
                            <?php
                                $services_selected = json_decode($row['services_selected'], true);
                                if (!empty($services_selected)) {
                                    foreach ($services_selected as $s) {
                                        $sObj = json_decode($s, true);
                                        if ($sObj) {
                                            echo htmlspecialchars($sObj['name']). " (".$sObj['price'].")<br>";
                                        }
```

```
<main class="main-content">
    <div class="table-wrapper">
        <div class="table-container">
            <table class="styled-table" id="appointmentsTable">
                <tbody>
                    <td><?= $row['reg_number']; ?></td>
                    <td><?= $row['appointment_date']; ?></td>
                    <td><?= $row['appointment_time']; ?></td>
                    <td>
                        <?php
                            $services_selected = json_decode($row['services_selected'], true);
                            if (!empty($services_selected)) {
                                foreach ($services_selected as $s) {
                                    $sObj = json_decode($s, true);
                                    if ($sObj) {
                                        echo htmlspecialchars($sObj['name'])." (".$sObj['price'].")<br>";
                                    }
                                }
                            }
                        ?>
                    </td>
                    <td><?= ucfirst($row['status']); ?></td>
                    <td><?= $row['created_at']; ?></td>
                </tr>
                <?php endwhile; ?>
                </tbody>
            </table>
        </div>
    </div>
```

Figure 3.3. 14 Customer Appointment Booking Source Code

3.3.8 Mechanic inventory usage

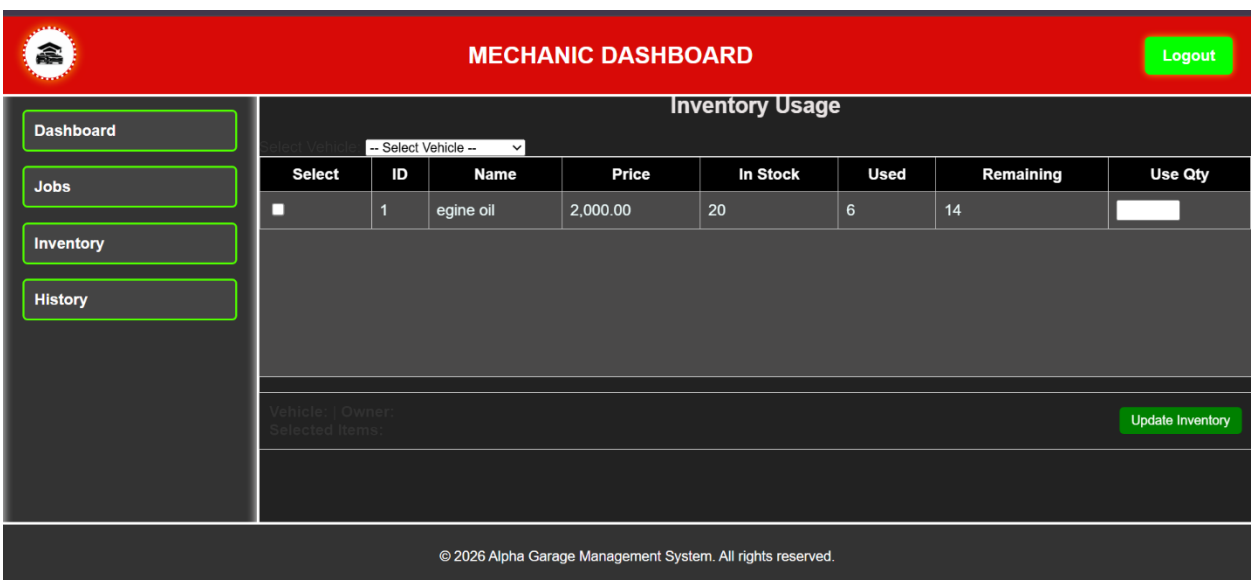


Figure 3.3. 15 Mechanic inventory usage

Mechanic inventory usage code:

```php
<main class="main-content">
    <h2 class="page-heading">
<?= $page_title; ?></h2>

    <!-- Popup Messages -->
    <?php include_once("../public/includes/popup.php"); ?>

    <!-- Add Item Form -->
    <form method="POST" class="inline-form">
        <input type="text" name="name" placeholder="Item Name" required>
        <input type="number" step="0.01" name="price" placeholder="Price" required>
        <input type="number" name="stock" placeholder="Initial Stock" required>
        <button type="submit" name="add_item" class="btn-green">Add Item</button>
    </form>

    <!-- Search Box (always stays here) -->
    <input type="text" id="searchBox" placeholder="Search items..." style="margin:15px 0; padding:5px; width:250px;">

    <!-- Update Item Form (appears below search bar when triggered) -->
    <form method="POST" class="inline-form" id="updateForm" style="margin-top:15px; display:none;">
        <input type="hidden" name="item_id" id="update_item_id">
        <input type="text" name="name" id="update_name" placeholder="Item Name" required>
        <input type="number" step="0.01" name="price" id="update_price" placeholder="Price" required>
        <button type="submit" name="update_item" class="btn-blue">Update Name/Price</button>
    </form>

    <!-- Inventory Table -->
    <div class="table-wrapper">
        <div class="table-container">
```

```php
<main class="main-content">
    <!-- Inventory Table -->
    <div class="table-wrapper">
        <div class="table-container">
            <table class="styled-table" id="inventoryTable">
                <thead>
                    <tr>
                        <th>ID</th><th>Name</th><th>Price</th><th>Initial Stock</th>
                        <th>Used</th><th>Remaining</th><th>Updated</th><th>Actions</th>
                    </tr>
                </thead>
                <tbody>
                    <?php while($row = $inventory->fetch_assoc()): ?>
                    <tr>
                        <td><?= $row['id']; ?></td>
                        <td><?= htmlspecialchars($row['item_name']); ?></td>
                        <td><?= number_format($row['price'],2); ?></td>
                        <td><?= $row['instock']; ?></td>
                        <td><?= $row['used']; ?></td>
                        <td><?= $row['remaining']; ?></td>
                        <td><?= $row['updated_at']; ?></td>
                        <td>
                            <!-- Select Item for Update -->
                            <button type="button" class="btn-orange"
                                onclick="selectItem(<?= $row['id']; ?>, '<?= htmlspecialchars($row['item_name']); ?>'
                                Edit Name/Price
                            </button>
                            <!-- Update Instock -->
                            <form method="POST" class="inline-form">
```

```
<main class="main-content">
    <div class="table-wrapper">
        <div class="table-container">
            <table class="styled-table" id="inventoryTable">
                <tbody>
                                onclick="selectItem(<?= $row['id']; ?>, '<?= htmlspecialchars($row['item_name']); ?>'
                                Edit Name/Price
                            </button>
                            <!-- Update Instock -->
                            <form method="POST" class="inline-form">
                                <input type="hidden" name="item_id" value="<?= $row['id']; ?>">
                                <input type="number" name="new_stock" placeholder="Add Instock" required>
                                <button type="submit" name="update_instock" class="btn-green">Update Instock</button>
                            </form>
                            <!-- Update Used -->
                            <form method="POST" class="inline-form">
                                <input type="hidden" name="item_id" value="<?= $row['id']; ?>">
                                <input type="number" name="add_used" placeholder="Add Used" required>
                                <button type="submit" name="update_used" class="btn-orange">Update Used</button>
                            </form>
                        </td>
                    </tr>
                    <?php endwhile; ?>
                </tbody>
            </table>
        </div>
    </div>
</main>
```

Figure 3.3. 16 Mechanic Inventory Usage Source Code

3.3.9 Receptionist vehicle registration

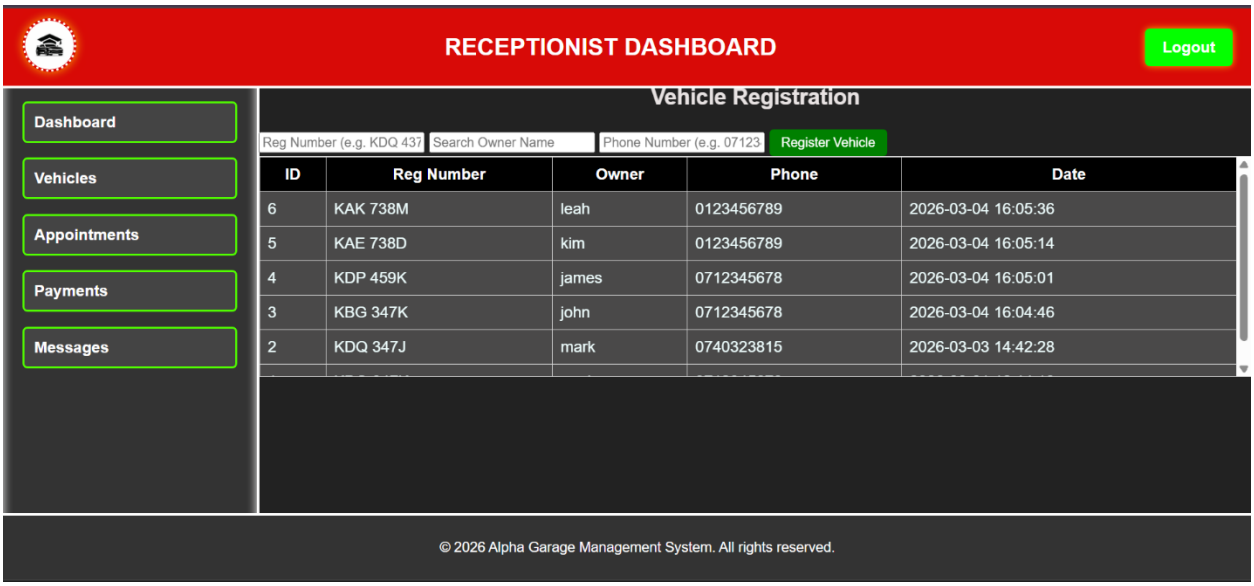


Figure 3.3. 17 Receptionist vehicle Registration

Receptionist vehicle registration source Code:

```
<main class="main-content">
    <h2 class="page-heading"><?= $page_title; ?></h2>
    <?php include_once("../public/includes/popup.php"); ?>

    <!-- Register Vehicle Form -->
    <form method="POST" class="inline-form">
        <input type="text" name="reg_number" placeholder="Reg Number (e.g. KDQ 437L)" required>

        <!-- Searchable input + dropdown for customer names -->
        <input list="customerList" name="owner_id" placeholder="Search Owner Name" required>
        <datalist id="customerList">
            <?php while($c = $customers->fetch_assoc()): ?>
                <option value="<?= $c['id']; ?>"><?= htmlspecialchars($c['username']); ?></option>
            <?php endwhile; ?>
        </datalist>

        <input type="text" name="phone" placeholder="Phone Number (e.g. 0712345678)" required>
        <button type="submit" name="register_vehicle" class="btn-green">Register Vehicle</button>

        <!-- Inline Search beside Register Vehicle button -->
        <input type="text" id="searchVehicles" placeholder="Search vehicles...">
    </form>

    <!-- Vehicles Table -->
    <div class="table-wrapper">
        <div class="table-container">
            <table class="styled-table" id="vehiclesTable">
                <thead>
                    <tr><th>ID</th><th>Reg Number</th><th>Owner</th><th>Phone</th><th>Date</th></tr>
```

```
<main class="main-content">
    <div class="table-wrapper">
        <div class="table-container">
            <table class="styled-table" id="vehiclesTable">
                <tbody>
                    <td><?= $row['id']; ?></td>
                    <td><?= htmlspecialchars($row['reg_number']); ?></td>
                    <td><?= htmlspecialchars($row['owner']); ?></td>
                    <td><?= htmlspecialchars($row['phone']); ?></td>
                    <td><?= $row['created_at']; ?></td>
                </tr>
                <?php endwhile; ?>
                </tbody>
            </table>
        </div>
    </div>
</main>

<script src="../public/js/global.js"></script>
<script>
// Search functionality for Vehicles
document.getElementById('searchVehicles').addEventListener('keyup', function() {
    let filter = this.value.toLowerCase();
    let rows = document.querySelectorAll('#vehiclesTable tbody tr');
    rows.forEach(row => {
        let text = row.textContent.toLowerCase();
        row.style.display = text.includes(filter) ? '' : 'none';
    });
});
```

Figure 3.3. 18 Receptionist Vehicle Registration Source Codes

3.4 Testing

Testing was conducted to ensure that all components of the system function correctly and meet user requirements.

3.4.1 Testing Activities Included:

i. **Unit Testing:** Individual components such as login and registration were tested separately.
ii. **Integration Testing:** Ensured that UI, logic, and database work together correctly.
iii. **User Testing:** Tested system usability and user experience.

Issues Identified and Fixes:

Issue	Cause	Solution
Login failure	Incorrect query	Fixed SQL query
Form not submitting	Missing validation	Added input validation
Slow loading	Inefficient queries	Optimized database queries

Table 3.4.1 Issues and Fixes

3.5 Deployment

Web-Based System Deployment

The system was deployed using infinity free web hosting platform.

Steps followed:

1. Prepared project files.
2. Uploaded files to the hosting server.
3. Configured database on the server.
4. Connected system to the live database.
5. Tested the system online.

Live System Link:

My web page url link is: alpha-auto-garage.infinityfreeapp.com

The system can now be accessed by users through a web browser.

CHAPTER FOUR: CONCLUSION AND RECOMMENDATION

4.1 Conclusion

This project focuses on developing a Garage Management System. Chapter One identifies problems in manual garage systems and sets out objectives such as improving booking, client management, and inventory tracking. Chapter two presents system designs, including user interfaces and flowcharts, to show how the system will work. Chapter three focuses on implementing the system using technologies like PHP, MySQL, HTML, CSS, and JavaScript.

During the project, I learned how to design and build a complete system, connect databases, and solve coding problems. Challenges faced included debugging errors, database connections, secure authentication ensuring proper system data validation, and managing time effectively.

4.2 Recommendation

Future developers can consider adding SMS or email notifications for appointments and integrating M-Pesa or card payments for easier transactions, improve staff attendance by using biometrics or QR codes instead of manual entry and adding a physical receipt printer for customers who prefer printed receipts. Additionally improve security with two-factor authentication, and make the system mobile-friendly or develop an app. These were not implemented due to limited time, budget, and technical constraints but would greatly improve the system.

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